Extended Security Analysis of the Alaniz Cipher

Chosen-Plaintext Attacks, Key Recovery, and Structural Properties of the Local Encryption Map

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Abstract

This document reports seven reproducible security experiments (Experiments 07–13) that complement and stress-test the security claims of the Alaniz Cipher (SHEC). The experiments probe the local encryption map $g_B(y) = y + B\,\sigma(y)$ for collision properties, attempt key recovery under known- and chosen-plaintext models, and analyse the algebraic structure exploitable by an adversary.

The central finding is contained in **Experiment 10**: both supported nonlinear maps ($\sigma = x^3$ and $\sigma = x^{p-2}$) satisfy a componentwise scaling law $\sigma(\lambda y)_i = \lambda^e\,\sigma(y)_i$, which allows a chosen-plaintext attacker to recover the per-node key matrices (A_v, B_v) using only $O(d)$ encryption queries—independently of the field size p and the size of $\mathrm{GL}(d, \mathbb{F}_p)$. This result directly contradicts the CPA-resistance claim in the original paper and is not addressed by the NL-SMIP hardness reduction, which concerns a distinct adversarial model. The remaining experiments characterise the collision structure of g_B (Experiments 07 and 11), measure the concrete complexity of exhaustive and differential key recovery under the known-plaintext model (Experiments 08, 09, 12, and 13), and provide the algebraic ground-truth necessary for evaluating any proposed fix.

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1 Introduction

The Alaniz Cipher encrypts a message encoded as a global section $s \in H^0(G, \mathcal{F})$ of a cellular sheaf over a graph G. At each node v the local ciphertext is

$$c_v = A_v \cdot s_v + B_v \cdot \sigma(A_v \cdot s_v), \qquad (A_v, B_v) \in \mathrm{GL}(d, \mathbb{F}_p)^2, \tag{1}$$

where $\sigma : \mathbb{F}_p^d \to \mathbb{F}_p^d$ is a public componentwise nonlinear map. The original paper argues that this construction resists chosen-plaintext attacks (CPA) because σ defeats linearisation: any attempt to model c_v as a linear function of s_v produces an inconsistent system as soon as enough ciphertext pairs are collected.

The experiments in this document examine whether non-linear attacks succeed. They are organised in order of increasing algebraic sophistication:

- **Experiments 07 and 11** characterise the map $g_B(y) = y + B\,\sigma(y)$, the core function the attacker must invert.
- **Experiment 08** establishes the exact threshold at which the key becomes trivially recoverable ($d = 1$) and confirms algebraically that linear attacks fail for $d \geq 2$.
- **Experiments 09 and 12** measure the empirical cost of key recovery under the known-plaintext model via exhaustive A-search.
- **Experiment 10** presents a direct CPA break requiring only $O(d)$ queries.
- **Experiment 13** proposes a differential column attack that recovers the columns of A independently, reducing the search space from $|\mathrm{GL}(d, p)|$ to $d\,(p^d - 1)$ candidates.

All experiments use the production implementation in `alaniz/` without modification. Random seeds are fixed for full reproducibility.

2 Protocol Notation

For reference, Table 1 summarises the quantities used throughout this document.

Decryption solves $y + B_r\,\sigma(y) = c_r$ at the root node, propagates via restriction maps, and uses cohomological filtering to select the valid global section.

3 Experiment 07 — Collision Table for the Local Encryption Map

3.1 Objective

Quantify the surjectivity and preimage multiplicity of g_B over the full domain $\mathbb{F}_p^d$ for $d \in \{1, 2, 4\}$ and $p \in \{17, 23, 29, 47, 59, 89, 131, 251\}$. This provides concrete evidence of whether g_B is injective, which determines decryption ambiguity.

Table 1: Notation used in this analysis.

Symbol	Meaning
p	Field characteristic (prime)
d	Fibre dimension
G	Graph with n nodes
$H^0(G, \mathcal{F})$	Space of global sections (plaintext space)
$s_v \in \mathbb{F}_p^d$	Local plaintext at node v
(A_v, B_v)	Secret key at node v; both in $\mathrm{GL}(d, \mathbb{F}_p)$
σ	Public componentwise map: `inverse` ($x_i \mapsto x_i^{p-2}$) or `cube` ($x_i \mapsto x_i^3$)
$y_v = A_v s_v$	Secret intermediate vector at node v
$g_B(y)$	$y + B\,\sigma(y)$: local encryption function

3.2 Methodology

For $d \in \{1, 2\}$, all p^d inputs are enumerated exactly. The preimage map $\{c \mapsto g_B^{-1}(c)\}$ is constructed and the following metrics are computed per random $B \in \mathrm{GL}(d, \mathbb{F}_p)$ (averaged over eight independent draws):

$$\text{image fraction} = \frac{|\mathrm{Im}(g_B)|}{p^d}, \qquad \gamma_B = \sum_{c \in \mathbb{F}_p^d} \left(\frac{|g_B^{-1}(c)|}{p^d} \right)^2. \tag{2}$$

For $d = 4$, where exhaustive enumeration is infeasible, a birthday probe with $3\lceil\sqrt{p^d}\rceil$ random samples guarantees a collision with high probability by the birthday bound. Every collision example is verified against the algebraic condition $\Delta y = B\big(\sigma(y_2) - \sigma(y_1)\big)$.

3.3 Findings

Across all tested primes and both σ functions, the image fraction stabilises near 0.53 for $\sigma =$ `inverse` and 0.66 for $\sigma =$ `cube` in dimension $d = 1$, and near 0.62–0.64 for $d = 2$. The map g_B is therefore consistently non-surjective: roughly one third of ciphertexts at each node are unreachable under a fixed key. Collisions are confirmed at all parameter sizes, including $d = 4$, $p = 251$.

3.4 Security Implications

Non-surjectivity implies two consequences. First, decryption at the root node admits multiple preimages (at most the maximum multiplicity, observed at 2–5 for $d = 2$), introducing ambiguity that the cohomological filter resolves. Second, the collision probability γ_B provides a direct empirical estimate of the parameter γ from Theorem 7.1 of the original paper.

4 Experiment 08 — Key Recovery at $d = 1$ and Linear Attack Diagnostics

4.1 Objective

Establish the exact threshold below which the key is algebraically recoverable, and confirm that the linearisation defence holds for $d \geq 2$.

4.2 Exact Recovery for $d = 1$

For $d = 1$ the encryption reduces to a scalar equation. Given two plaintext–ciphertext pairs (s_1, c_1) and (s_2, c_2):

Case $\sigma =$ `inverse`. Multiplying $c_i = as_i + b(as_i)^{-1}$ through by as_i and subtracting the two equations yields:

$$a = \frac{c_1 s_1 - c_2 s_2}{s_1^2 - s_2^2}, \qquad b = c_1(as_1) - (as_1)^2. \tag{3}$$

Case $\sigma =$ `cube`. The two equations form a 2×2 Vandermonde-type system in $(a,\, ba^3)$, solved by Cramer's rule with determinant $s_1 s_2^3 - s_2 s_1^3$.

Both formulas are exact over $\mathbb{F}_p$ and require only two captures.

4.3 Linear Surrogate Failure for $d \geq 2$

For $d \in \{2, 4\}$, the experiment builds the linear surrogate $c_v = Ms_v$ (treating the key as a single $d \times d$ matrix M) and tracks its rank and consistency as captures accumulate. As expected, the nonlinearity of σ renders the system inconsistent after enough pairs, confirming that pure linearisation fails.

4.4 Findings and Security Implications

Dimension $d = 1$ provides **no computational security**: the key is recoverable with 2 plaintext-ciphertext pairs using only modular arithmetic. The minimum secure dimension is $d \geq 2$. However, Experiment 10 shows that a non-linear CPA attack succeeds at all dimensions.

5 Experiment 09 — Exhaustive Known-Plaintext Key Recovery

5.1 Objective

Measure the concrete cost of recovering the per-node key (A_v, B_v) under the known-plaintext model via exhaustive search over $A \in \mathrm{GL}(d, \mathbb{F}_p)$ with linear solve for B.

5.2 Methodology

For each candidate matrix $\hat{A}$, compute $y_i = \hat{A}\, s_i$ and $z_i = \sigma(y_i)$ for d fit captures, then solve the linear system

$$c_i - y_i = B\, z_i, \qquad i = 1, \ldots, d, \tag{4}$$

for the d^2 entries of B. Validate the recovered $(\hat{A}, \hat{B})$ against all k captures. The campaign covers $p \in \{17, 23, 29, 47, 59, 89, 131, 251\}$, $d = 2$, with $k \in \{4, 8, 12\}$ captures. For $|\mathrm{GL}(2, p)| \leq 10^5$ exhaustive enumeration is used; for larger primes 5 000 random samples are drawn.

5.3 Findings and Security Implications

For $p \leq 29$, $|\mathrm{GL}(2, p)| \leq 7 \times 10^5$, and the true key is recovered exactly in all trials. For $p \geq 47$ the sample probability is negligible ($\approx 5000/|\mathrm{GL}(2, p)|$). The number of false positives drops sharply with k: at $k = 12$ no false positive survives any exhaustive run. The exhaustive A-search is intractable at standard parameters ($d = 4$, $p \approx 2^{31}$), but Experiment 10 renders this bound irrelevant by recovering the key in $O(d)$ chosen-plaintext queries regardless of p.

6 Experiment 10 — Direct CPA Key Recovery via Scaling Homogeneity

This is the main security finding of this analysis.

6.1 Objective

Demonstrate a chosen-plaintext attack that recovers (A_v, B_v) in $O(d)$ encryption queries, without any exhaustive search over $\mathrm{GL}(d, \mathbb{F}_p)$.

6.2 Mathematical Basis

Both supported nonlinear maps satisfy a **componentwise scaling law**:

$$\sigma(\lambda\, y)_i = \lambda^e\, \sigma(y)_i \qquad \forall\, y \in \mathbb{F}_p^d,\ \lambda \in \mathbb{F}_p^*, \tag{5}$$

with $e = 3$ for $\sigma = \texttt{cube}$ and $e \equiv p - 2 \pmod{p - 1}$ for $\sigma = \texttt{inverse}$.

Because B acts linearly on $\sigma(y)$, equation (5) extends to the full ciphertext. For plaintexts $s = t\, e_j$ (scalar multiples of the j-th canonical basis vector of the H^0-coefficient space), the local ciphertext satisfies

$$c(t\, e_j) = t\, a_j + t^e\, B\, \sigma(a_j), \tag{6}$$

where $a_j = A\, e_j$ is the j-th column of A. Querying $s_1 = e_j$ $(t = 1)$ and $s_2 = \lambda\, e_j$ $(t = \lambda)$ and writing $w_j = B\, \sigma(a_j)$ gives

$$\begin{pmatrix} 1 & 1 \\ \lambda & \lambda^e \end{pmatrix} \begin{pmatrix} a_j \\ w_j \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}. \tag{7}$$

The determinant is $\Delta = \lambda^e - \lambda$. Choosing any λ with $\Delta \neq 0$ (which exists for all $p > 3$) yields a *closed-form* solution:

$$a_j = \frac{\lambda^e c_1 - c_2}{\Delta}, \qquad w_j = \frac{-\lambda c_1 + c_2}{\Delta}. \tag{8}$$

Repeating for $j = 0, \ldots, d-1$ recovers the effective key matrix $A_{\text{eff}} := A\,P$ in $2d$ queries, where P is the publicly computable coefficient-to-stalk map at the root node. Recovering A in stalk coordinates:

$$A = A_{\text{eff}} \cdot P^{-1}. \tag{9}$$

With A_{eff} known, B is recovered by collecting d additional queries with linearly independent $z^{(i)} = \sigma(A_{\text{eff}}\, s^{(i)})$ and solving

$$B = W\,Z^{-1}, \qquad W_i = c^{(i)} - A_{\text{eff}}\, s^{(i)}, \quad Z = \left[z^{(1)} \cdots z^{(d)}\right]. \tag{10}$$

Total query count: $2d + d = 3d$ (worst case), or $2d + 1$ when the basis queries already provide an invertible Z.

6.3 Experimental Validation

The attack is executed against all primes $p \in \{17, 23, 29, 47, 59, 89, 131, 251\}$, both σ functions, and dimensions $d \in \{2, 4\}$. In every case:

- A_{eff} is recovered exactly (`A_eff_match = True`).
- A in stalk coordinates is recovered exactly (`A_stalk_match = True`).
- B is recovered exactly (`B_match = True`).
- The recovered key passes 20 fresh validation queries (`valid = True`).

For $d = 4$ the total number of encryption queries used is 12, independently of p.

6.4 Security Implications

This result has the following consequences for the security claims of the original paper:

1. **The CPA-resistance claim is falsified for all parameter sets.** The attack does not require linearisation and is not blocked by the nonlinearity of σ. On the contrary, the scaling homogeneity of σ *is* the algebraic property that enables the attack. The statement "linearisation fails because $B_v\, \sigma(A_v s_v)$ mixes key and message nonlinearly" remains true but does not capture this attack vector.

2. **The NL-SMIP hardness reduction is not affected.** The reduction from MQ establishes that recovering the key from a polynomial system with secret coefficients is NP-hard. The scaling attack does not solve NL-SMIP; it avoids it entirely by exploiting a structural property of σ outside the polynomial formulation.

3. **Post-quantum claims are unaffected in substance.** The scaling attack is purely algebraic and applies with the same $O(d)$ complexity classically and quantumly. It does not exploit a quantum speedup; it is simply faster than any exhaustive search.

4. **The attack extends to all nodes simultaneously.** Each node's key (A_v, B_v) is independent and the attack is node-local. Querying all n nodes requires $3dn$ encryption queries. For a 6-node path with $d = 4$ this is 72 queries.

5. **The attack is repairable.** Any σ satisfying $\sigma(\lambda y) \neq \lambda^e \sigma(y)$ for all e would defeat this specific attack. Extending the σ-selection analysis of Experiment 04 to include scaling homogeneity is the natural direction for a fix.

7 Experiment 11 — Statistical Collision Study of g_B

7.1 Objective

Obtain statistically robust estimates of the collision properties of g_B across the full parameter space, extending the spot-check of Experiment 07 to large samples of B matrices.

7.2 Methodology

For $d = 2$ and all primes, exact enumeration over p^d inputs is repeated for 12–24 independent draws of $B \in \mathrm{GL}(d, \mathbb{F}_p)$. For $d = 4$, 30 000 random inputs are sampled per B. Per configuration the following are reported:

- Mean image fraction $\pm$ standard deviation over the B-sample.
- Mean preimage multiplicity $p^d/|\mathrm{Im}(g_B)|$.
- Worst-B examples (lowest image fraction).
- Aggregated multiplicity histogram.

7.3 Findings and Security Implications

The image fraction for $\sigma = \texttt{inverse}$ stabilises near 0.63 for $d = 2$ with very low variance ($\sigma_{\mathrm{std}} < 0.01$ for $p \geq 59$), confirming that non-surjectivity is a stable property of σ, not an artefact of unlucky B choices. The maximum preimage multiplicity remains bounded below 7 for $d = 2$, $p \leq 251$, consistent with Theorem 7.1 of the original paper. Together with Experiment 07, this experiment provides empirical estimates of $\gamma \approx 1 - 0.63 = 0.37$ and $L \approx 1/0.63 \approx 1.59$ for the $\sigma = \texttt{inverse}$, $d = 2$ configuration.

8 Experiment 12 — Key Recovery Diagnostics and Minimum Capture Analysis

8.1 Objective

Measure the minimum number of known-plaintext pairs required for unique key recovery at $d = 2$ (exhaustive) and $d = 3$ (sampled), and track how quickly false positives are eliminated as captures accumulate.

8.2 Methodology

For each $\hat{A}$ candidate (exhaustive for $d = 2$, $p = 17$; 60 000 random samples for $d = 3$), B is solved from the first d captures and the resulting candidate is tested against an increasing prefix of up to 20 captures. The *first failure index*—the smallest k at which the candidate fails—is recorded. The minimum k at which exactly one candidate (the true key) remains is denoted $k_{\min}$.

8.3 Findings and Security Implications

For $d = 2$, $p = 17$: the number of surviving candidates drops from thousands at $k = d = 2$ to single digits by $k = 6$, and to exactly 1 (no false positives) by $k \approx 10$–15, depending on σ. For $d = 3$ in sampled mode, unique recovery is achieved by $k \approx 12$–18 captures. Each capture carries roughly $d^2 \log_2 p$ bits of key information, consistent with the polynomial system structure. These bounds are primarily relevant for toy configurations; at standard parameters the exhaustive A-search is infeasible (Experiment 09).

9 Experiment 13 — Differential Column Attack

9.1 Objective

Test whether the columns of A can be recovered *independently* via chosen-plaintext differential queries along coordinate axes, and measure the resulting reduction in search complexity.

9.2 Mathematical Basis

For plaintexts $s = t\, e_j$, consecutive ciphertext differences are:

$$\Delta c_t = c\Big((t+1)e_j\Big) - c\Big(t\, e_j\Big) = a_j + B\Big(\sigma\Big((t+1)a_j\Big) - \sigma(t\, a_j)\Big). \tag{11}$$

For a guessed column $\hat{a}_j$, this is a linear equation in B. Collecting d such differentials and denoting $\Delta\sigma_t = \sigma((t+1)\hat{a}_j) - \sigma(t\hat{a}_j)$:

$$\Delta c_t - \hat{a}_j = B\, \Delta\sigma_t. \tag{12}$$

If the matrix $[\Delta\sigma_0 \cdots \Delta\sigma_{d-1}]$ is invertible, B is determined uniquely. Each column of A is recoverable by searching $p^d - 1$ nonzero candidates for $\hat{a}_j$, reducing the total search space from $|\mathrm{GL}(d,p)| \approx p^{d^2}$ to $d\,(p^d - 1)$.

9.3 Findings

$d = 2$, $p = 17$: $p^2 - 1 = 288$ candidates per axis *vs.* $|\mathrm{GL}(2, 17)| = 69\,360$ full matrices. Unique recovery in 4–6 differentials in all tested configurations.

$d = 3$, $p = 17$: $p^3 - 1 = 4\,912$ candidates *vs.* $|\mathrm{GL}(3, 17)| \approx 1.4 \times 10^7$. Unique recovery in 4–7 differentials.

Limitation. If column a_j has a zero in component i, then $\sigma(t\, a_j)_i = 0$ for all t, making the i-th row of $\Delta\sigma$ identically zero. The corresponding entries of B remain undetermined and the attack produces false negatives for those configurations.

9.4 Security Implications

The differential column attack reduces per-column search from p^{d^2} to $p^d - 1$ candidates—an exponential improvement for $d \geq 2$. It is dominated by the $O(d)$-query attack of Experiment 10 in practical terms, but provides an independent recovery path useful when direct query access is restricted.

10 Discussion

10.1 The Scaling Homogeneity as a Structural Vulnerability

The root cause of the CPA break in Experiment 10 is the interplay between three design choices:

1. σ is applied *componentwise*, enabling $\sigma(\lambda y)_i = \lambda^e \sigma(y)_i$.
2. B acts *linearly* on $\sigma(y)$, so $B\,\sigma(\lambda y) = \lambda^e\, B\,\sigma(y)$.
3. The plaintext space includes *scalar multiples* of any vector, allowing the attacker to query $(s, \lambda s)$ and decouple $y = As$ from $w = B\sigma(As)$ via a 2×2 linear system.

Any two of these three properties in isolation would not be sufficient for the attack. A fix must therefore break at least one of them. The most natural option is replacing σ with a map that lacks componentwise scaling while preserving bijectivity and compatibility with decryption.

10.2 Relationship to the Original Security Proof

The original paper's CPA resistance argument considers designs where σ is applied before or independently of the secret map, showing that in those cases the key can be recovered linearly. The final design places σ *after* the secret linear map A_v, which does defeat linearisation. Experiment 10 shows that defeating linearisation is necessary but not sufficient for CPA security: the scaling homogeneity provides an alternative attack that does not require the attacker to linearise the system.

The NL-SMIP hardness proof (Lemma 6.2 of the original paper) establishes that recovering the key from a polynomial system with secret coefficients is NP-hard. The

scaling attack models a different scenario—one in which the attacker controls the plaintexts and exploits algebraic structure in σ. These two models are not equivalent, and the NP-hardness of NL-SMIP does not imply CPA security.

10.3 Collision Properties and Decryption Correctness

Experiments 07 and 11 together establish that g_B is non-surjective with image fraction ≈ 0.63 for $d = 2$. This provides empirical estimates of $\gamma \approx 0.37$ and $L \approx 1.59$ for the correctness bound in Theorem 7.1 ($\Pr[\text{ambiguity}] \leq (L-1)\,\gamma^{n-1}$). For graphs with $n \geq 6$ nodes, decryption ambiguity is negligible in practice.

11 Conclusion

This analysis presents seven reproducible security experiments that probe the Alaniz Cipher from multiple adversarial perspectives. The key conclusions are:

1. **Dimension $d = 1$ is trivially insecure**: the key is algebraically recoverable from 2 plaintext-ciphertext pairs (Experiment 08).
2. **The local map g_B is consistently non-surjective** (image fraction ≈ 0.63) with bounded preimage multiplicity, validating the decryption uniqueness analysis (Experiments 07 and 11).
3. **Known-plaintext exhaustive key recovery** is feasible for toy parameters ($d = 2$, $p \leq 29$) and requires approximately 10–15 captures for unique recovery; it is infeasible at proposed standard parameters (Experiments 09 and 12).
4. **A direct CPA break exists for all parameters**: the scaling homogeneity $\sigma(\lambda y) = \lambda^e\,\sigma(y)$ allows recovery of the per-node key (A_v, B_v) in $O(d)$ chosen-plaintext queries, independently of p (Experiment 10). This is the principal security finding of this analysis.
5. **A differential column attack** reduces the per-column search space from $|\mathrm{GL}(d,p)|$ to $d\,(p^d - 1)$ candidates (Experiment 13).

The scaling attack of Experiment 10 is a significant vulnerability that requires attention before the scheme can be considered secure under standard CPA definitions. The natural direction for a repair is to replace the componentwise scaling-homogeneous σ with a map that mixes components across the d-dimensional fibre while preserving the algebraic properties required for efficient decryption.

Experiment Summary

Experiments 07–13 are implemented in `experiments/07_collision_table.py` *through* `experiments/13_differential_column_attack.py`*. All random seeds are fixed; results are fully reproducible.*

Table 2: Summary of Experiments 07–13.

Exp	Model	Attack / Analysis	Complexity	Main Result
07	—	g_B collision characterisation	$O(p^d \cdot b)$	Image fraction $\approx$ 0.63; collisions universal
08	KPA	Algebraic key recovery	$O(1)$ for d=1	d=1 trivially broken; linearisation fails at $d\geq 2$
09	KPA	Exhaustive A-search + linear B	$O(\lvert\mathrm{GL}(d,p)\rvert)$	Feasible for $p \leq 29$; intractable at standard params
10	**CPA**	**Scaling homogeneity attack**	$\boldsymbol{O(d)}$ **queries**	**Full key recovery at all params; CPA claim falsified**
11	—	g_B statistical multiplicity	$O(p^d \cdot b)$	Stable image fraction; worst-B families identified
12	KPA	Capture count diagnostics	$O(\lvert\mathrm{GL}(d,p)\rvert \cdot k)$	Unique recovery in $\approx$ 10–15 captures for d=2
13	CPA	Differential column attack	$O(d(p^d-1))$	Search from p^{d^2} reduced to p^d-1 per column

KPA = Known-Plaintext Attack; CPA = Chosen-Plaintext Attack.